

# Project Proposal

*Introduction to Algorithms and Software Programming (SL 203)*

## Optimizing Cardiac Ward Workflows: An Algorithmic Approach to Dynamic Task Management and Allocation

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### 1. Objective:

This project aims to design and build an algorithm-based ward task management system that optimally schedules tasks to healthcare workers based on their skills, availability, and priority. This project focuses specifically on the cardiology ward, as a scoped implementation for a course project. The expected outcome is a demonstrably more optimised task allocation system that can be used in hospitals to improve the quality of care, minimize task turnaround time, and reduce the workload of healthcare workers.

### 2. Rationale:

Healthcare professionals have faced burnout for decades. Many hospitals in India are still using pen and paper to allocate tasks to healthcare workers. Doing so results in inefficient resource utilization. For example, the first task on the paper gets done first, irrespective of its priority.

### 3. Approach:

The system will be built using Kahn's Topological Sort and greedy algorithms. Kahn's algorithm will be used to get the correct execution order (e.g., Troponin + ECHO → Stress Test). Greedy algorithms for priority mapping (e.g., suspected MI → stable chest pain → routine follow-up). The system will be improved iteratively after doing clinical validation and sanity check. Explore the possibility of using other algorithms and data structures. Perform benchmarking against the standard pen and paper method.

### 4. Timeline:

- **Week 1:** Obtain a list of cardiovascular diseases and tests (From Mayo Clinic); Filter for diseases and tests that requires admission; Identify suitable data structures for patients, healthcare workers, diseases, tests, and their relationships.
- **Week 2:** Implement Kahn's Topological Sort and greedy algorithms; perform initial clinical validation and sanity checks.
- **Week 3:** Iteratively refine the system based on clinical evaluation; Benchmark this over the standard pen and paper method. finalize the system and prepare the presentation.

### 5. Possible Issues

**1:** Pen-and-paper method does first-come, first-served. Here urgent patients may be delayed if staff are occupied with low-priority tasks. Implementing a priority-based scheduling algorithm in a way that hospitals can easily adapt and use it on the Workflows will aid here. **Example:** While i was working as a doctor, I observed a stable patient with chest pain being deprioritised because their initial ECG was normal. As he was having the chest pain for a week, CT angiogram was performed as the patient insisted (According to the guideline when the ECG is normal CT angiogram is need). But the CT showed a 90% blockage requiring emergency surgery. **2:** Lack of optimized data structures and architectural design makes retrieval and updates delay adoption from the users. Using hash maps for quick lookups and priority queues for task scheduling to reduce latency would help in this case so that healthcare workers can use the system easily **3:** Government hospitals still serve more than half the population but remain under-resourced with no standard digital entry point for EHR data. Designing the system to be modular and adaptable making healthcare workers access it directly from their phone will help in this case. *These issues form the core focus of this project.*